

Enhancing cross-domain recommendations: Leveraging personality-based transfer learning with probabilistic matrix factorization

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ABSTRACT

The conventional method of computing personality scores through extensive questionnaire-based surveys poses practical challenges in real-world scenarios. An alternate route is to predict personality scores from user reviews by analysing various linguistic features such as writing style, word choices, and specific phrases. However, the reviews are domain-dependent and classification models trained on one domain cannot be readily applied to other domains. To mitigate this challenge, we propose a cross-domain recommendation framework called PEMF-CD which leverages a novel mixing strategy to integrate user reviews from multiple domains with common joint embedding space and predict user personality scores using a transformer model. By capturing the underlying semantics and latent representations within the textual data, the transformer architecture can effectively model the linguistic cues to infer users' personality traits, and the learning is transferred across domains. To further enhance the recommendation process, our model integrates personality-wise and rating pattern-based similarities of users into a probabilistic matrix factorization method that fosters user neighbourhoods based on similarity scores among users. Comprehensive experiments were conducted using five real-world datasets from TripAdvisor and Amazon with varied numbers of users, items, and reviews of up to 44,187, 26,386, and 426,791, respectively. The performance has been benchmarked against thirteen baseline algorithms and the experimental results demonstrate a significant improvements of up to 24.72%, 64.28%, 48.79%, and 61% in RMSE, and 55.9%, 76.7%, 67.6%, and 71.5% in MAE for a 90:10 train-test split with Digital Music, Fashion, Magazine Subscriptions and Video Games datasets from Amazon, respectively. Similar results have been observed for the 80:20 train-test split.

1. Introduction

The advent of Web 2.0 has generated a revolutionary shift in online communication, facilitating the rise of social networks, blogs, video-sharing sites, e-commerce platforms, etc. This dynamic landscape has witnessed a rapid growth in user-generated content, which presents an opportunity to uncover valuable business insights. However, the sheer volume of data poses a significant challenge when it comes to searching for relevant information. On this ground, efficient recommendation systems are highly desirable for filtering and providing information according to the preferences of the users. Traditionally, collaborative filtering (Schafer, Frankowski, Herlocker, & Sen, 2007) and content-based filtering approaches (Pazzani & Billsus, 2007) are widely used, where the user and item features and their historical interactions are employed to generate the recommendations. While these methods offer computational simplicity and operational convenience, they are plagued by well-known issues and cold-start problems.

They struggle to generate recommendations in the absence of historical records (sparsity) or when encountering new users or items (cold-start). Additionally, these approaches tend to overlook the intricate dependencies between user and item features. A new stream of research using user personality features has emerged (Dhelim, Aung, Bouras, Ning, & Cambria, 2022), mainly because personality traits can serve as an independent source of information that can offer valuable insights. Furthermore, prior research has shown a significant correlation between user preferences and purchase behaviour (Finn, 1997; Kassarjian, 1971). However, obtaining user personality scores poses a significant challenge. Traditional methods rely on questionnaires or surveys (Kleć, Wiczorkowska, Szklanny, & Strus, 2023; Soto & John, 2017), which can be tedious, prone to bias, and yield inaccurate results. Additionally, conducting extensive surveys and collecting large sample sizes can be impractical to obtain users' personality (Hu & Pu, 2011). An alternative approach is to analyse user-generated content, such as posts from

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social media and reviews, to infer personality traits (Christodoulou, Gregoriades, Pampaka, & Herodotou, 2022; Sánchez-Fernández, Ruiz, & Jiménez, 2023). Psychological studies (Kaushal & Patwardhan, 2018; Tausczik & Pennebaker, 2010) have indicated that personality can be discerned from the writing style and the frequency of specific words or phrases used using machine learning techniques (Cachero, Rico-Juan, & Macià, 2022; Mehta, Majumder, Gelbukh, & Cambria, 2020). Text analysis tools, such as LIWC (Mairesse, Walker, Mehl, & Moore, 2007),¹ can extract personality features from the text. Nevertheless, the reviews are domain-dependent, and a classification model trained on one domain may not be readily applicable to another domain. Moreover, LIWC requires extensive configuration and may have different operational requirements for each domain.

In light of these challenges, there is a need for a cross-domain recommendation framework that can learn user-personality traits in one domain and predict personality in another. In this study, we propose a personality-based recommendation system called **PEMF** (Personality Enabled Matrix Factorization), which capitalizes the matrix factorization and incorporates personality-wise similarity between users. We create a neighbourhood of users based on personality-wise similarity, as previous research suggests that individuals with similar personalities tend to have similar preferences for items (Youyou, Stillwell, Schwartz, & Kosinski, 2017). We adopted the widely recognized Five-factor model (Big Five model) of personality (Goldberg, 1990), which categorizes personality into five traits: Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism. To validate our approach, we conducted experiments using the TripAdvisor dataset² (Roshchina, Cardiff, & Rosso, 2015), which includes Big Five personality scores of users who rated hotels and restaurants on a 1–5 scale, predicted using LIWC tool. Furthermore, we extend **PEMF** to a cross-domain framework called **PEMF-CD** (Personality Enabled Matrix Factorization for Cross Domain), which employs transfer learning techniques (Torrey & Shavlik, 2010) to learn personality scores in one domain and transfer them to another domain. Unlike traditional cross-domain recommenders that rely on common users or establish relationships between domains, **PEMF-CD** achieves this by jointly embedding user-generated reviews from both domains and learning personality scores using regression techniques, even in the absence of common users. These scores are then utilized in **PEMF-CD** to generate recommendations. We evaluated the performance of our proposed approaches against thirteen baseline algorithms, conducting extensive experiments using the TripAdvisor dataset and Amazon datasets³ (Ni, Li, & McAuley, 2019).

We have conducted extensive experiments to evaluate the efficacy of the proposed approaches and compared them against thirteen baseline algorithms. **PEMF** has been tested on a publicly available dataset from TripAdvisor showing a significant improvement up to 71% and 74.8% in RMSE and MAE, respectively. Further, **PEMF-CD** has been evaluated on four Amazon datasets and the results indicate an improvement of up to 24.72%, 64.28%, 48.79%, and 61% in RMSE, and 55.9%, 76.7%, 67.6%, and 71.5% in MAE for a 90:10 train–test split with Digital Music, Fashion, Magazine Subscriptions and Video Games datasets from Amazon, respectively. Similar results have been observed for the 80:20 train–test split.

Overall, in this research work, we aim to develop a novel personality-based cross-domain recommender system that considers similarity between users based on personality traits. The main contributions of this paper are as follows:

1. The study devises a method to acquire personality scores from user-generated reviews using transfer learning that overcomes the challenges of survey-based methods of collecting personality scores.
2. The study develops a personality-based recommender system that incorporates user personality features in the probabilistic matrix factorization technique to provide item recommendations with improved accuracy.
3. A cross-domain recommender system is designed that enables learning personality features from other domains when personality scores are not available.
4. The cross-domain framework adopts a novel mixing strategy and joint embedding of source and target domain texts for prediction.
5. Significance of personality features and effect of similar users in a neighbourhood have been demonstrated in the overall performance improvement of the recommender system.

The structure of the paper is as follows. Section 2 discusses the elaborated literature review with the research objectives. In Section 3, we provide a detailed description of Big Five personalities upon which our proposed work is based, followed by explanations regarding probabilistic matrix factorization for recommendation and existing models and approaches to enable a cross-domain framework and overcome its challenges. Section 4 presents our proposed algorithms, namely **PEMF** and **PEMF-CD**. After that, in Section 5, we present a brief discussion on the datasets used from TripAdvisor and Amazon. In Section 6, we experimentally validated the efficiency of our proposed approaches over thirteen state-of-the-art baseline models, out of which nine are specially for cross-domain recommendations and two with personality based recommendation in cross-domain framework. Section 7 discusses the impact of the personality features and user neighbourhood that owes to the recommendations and the theoretical and practical implications of our research work. This is followed by a conclusion in Section 8 summarizing the entire work along with highlighting the limitations and future research directions.

2. Literature review and research objectives

In this section, we discuss the prior works in the relevant fields of psychology and social science, discussing the dependency of personality and user preferences and how personality features can be predicted from the text. We also explore the present challenges and recent works in personality-based recommender systems.

2.1. Personality and its correlation on item preferences

The prior literature (Finn, 1997; Kassarjian, 1971) in psychology and marketing sciences has demonstrated the impact of an individual's personality on their preferences for various items. It also plays a role in determining user behaviour in online platforms (Al-Samarraie, Eldenfria, & Dawoud, 2017). Many marketing researchers have attempted to establish a connection between the purchase pattern, item choice, consumer behaviour with the user's personality. For instance, Kassarjian (1971) shed light on the impact of customer personality on consumer behaviour, highlighting the correlation between personality and preferences, which demonstrated a significant dependency between the two. Another study by Finn (1997) observed the relationship between media exposure and personality traits, specifically the Big Five personality traits, concerning preferences for TV shows, movies, and books. The findings revealed that individuals with higher levels of extraversion were more inclined towards action movies and less interested in reading books, whereas those with lower levels of extraversion showed a stronger preference for reading books and artistic movies. Notably, personality has been proven to be having a significant impact on user preferences in the music (Melchiorre, Zangerle, & Schedl, 2020) and

¹ Linguistic Inquiry and Word Count, <https://www.liwc.app/>

² TripAdvisor dataset 2015, https://www.researchgate.net/publication/301543515_TripAdvisor_dataset_with_personality_scores/link/5718059c08aed43f63220a27/download

³ Amazon dataset 2019, <https://nijianmo.github.io/amazon/index.html>

movie domains (Abolghasemi, Engelstad, Herrera-Viedma, & Yazidi, 2022; Khan, Mukta, Ali, & Mahmud, 2020). Kleč et al. (2023) identified the influence of personality in music and showed, using an experiment, how recommendations differ when different aspects of personality are considered. Polignano, Narducci, de Gemmis, and Semeraro (2021) used personality features obtained from social media posts to analyse the preferences of music. Similarly, in the realm of online shopping, Jia, Wang, Liu, and Jiang (2022) have established a noteworthy association between personality factors and brand choices on e-commerce platforms. Similarly, there have been works (Jie, Poulouva, Haider, & Sham, 2022; Tarka, Kukar-Kinney, & Harnish, 2022; Wang et al., 2022) which demonstrate the impacts of personality aspects on buying behaviours.

2.2. Personality-based recommender system

In the work by Hu and Pu (2011) developed the first personality-based recommender system. This has sparked the utilization of personality features in recommendations across various domains like movies (Abolghasemi et al., 2022; Khan et al., 2020), music (Martijn, Conati, & Verbert, 2022) and online marketing (Vatani, Rahmani, & Javadi, 2023) in recent years leading to an enhanced performance with quality recommendations.

Khan et al. (2020) experimented on users on a movie recommendation platform to unravel how diversity and popularity can be achieved by leveraging their personality traits. In the music domain, Martijn et al. (2022) added cognition with openness traits and showed how music recommenders can be designed to cater to users with higher and lower levels of the traits. Polignano et al. (2021) created an emotion-driven music recommender system by analysing personality from social media posts. While diversity in the recommendation is addressed, there has been limited exploration of leveraging user similarity based on personality traits. Abolghasemi et al. (2022) devised a movie recommender system targeted at a group of people with different personality traits. They aimed to incorporate the assertiveness of users and their influence on each other in driving the recommendations. Vatani et al. (2023) considered the personality aspects of users and their similarity to each other under different product categories. Additionally, incorporating other features such as demographic characteristics (Khodabandehlou, Hashemi Golpayegani, & Zivari Rahman, 2021), and even multiple personality models together (Aguilar, Fechine, & de Barros Costa, 2020) can further enhance the performance of the model.

Traditionally, early personality-based recommender systems relied on collecting personality scores through surveys and questionnaires (Hu & Pu, 2011). However, gathering a large sample size through surveys can be challenging and often leads to inaccuracies. In recent years, text mining techniques have emerged as an alternative approach, utilizing user-generated reviews to compute personality traits based on writing patterns, word choices, and expressiveness (Christodoulou et al., 2022; Roshchina et al., 2015; Yakhchi, Beheshti, Ghafari, & Orgun, 2020). This shift towards extracting personality from the text not only addresses the issue of data sparsity issue but also provides a way to infer personality traits without the need for explicit surveys, enabling more efficient and accurate personality-based recommendations.

Personality features, being independent of specific domains, can be utilized in developing cross-domain recommender systems. Fernández-Tobías and Cantador (2015) were pioneers in creating a cross-domain personality-based recommendation system. In 2016, the same authors (Fernández-Tobías, Braunhofer, Elahi, Ricci, & Cantador, 2016) introduced three approaches that associated personality features with the model for cross-domain recommendations. Further, Yakhchi et al. (2020) developed a cross-domain recommender system that leveraged the common users between two domains and their associated personality features and domain knowledge. They utilized data from social media platforms to extract user preferences and personality traits for the common users. However, finding common users across multiple domains is generally impractical. On the other hand, text mining tools

like LIWC, which can extract personality scores, often heavily rely on domain-specific configurations, making them expensive and less applicable across domains. In this regard, employing transfer learning methods to learn personality scores from another dataset that already contains such scores offers a more convenient approach to establishing a cross-domain framework for recommendation. Recently, Wang, Zuo, Li, and Wu (2021) demonstrated how transfer learning strategies could be adopted to establish connections between two domains through the joint embedding of text reviews, even in the absence of any common users between the domains. This helps largely in solving cold start issues when there is a new user coming having no item interaction previously. This issue is further addressed by Yu et al. (2021), proposing a cross-domain recommender system solving a bi-optimization problem using a selective ensemble technique and Soundariya, Manisekaran, Ramakrishnan, Ganesh, and Keerthi (2022), designing a cross-domain recommender using neural networks for getting personalized preferences along with diversity. Furthermore, Cao et al. (2023) makes recommendations for cold start users in any domain by using that domain's information combined with information transferred from other domains where the user has interacted. Table 1 summarizes the prior works on personality-based recommender systems.

2.3. Research objectives

Following the literature review, we formally outline the primary objectives of our research as follows:

- Development of a novel personality-based recommender system: Our goal is to create a novel recommender system that utilizes user personality features to provide personalized item recommendations, and with that, we aim to enhance the accuracy and relevance of the recommendations.
- Acquisition of personality scores from user-generated text content: We aim to overcome the challenges of bias and inaccuracy faced by the traditional methods of surveys to collect personality data by devising a method that extracts personality scores directly from the text content of user-generated reviews. This would analyse the writing style, word choice, and frequency of specific phrases to infer personality traits without relying on explicit surveys.
- Design of a cross-domain recommender system: We intend to develop a novel cross-domain recommender system that leverages information from other domains to learn personality features when explicit personality scores are not available. This approach will enable us to overcome data sparsity issues and provide personalized recommendations even in domains with limited user-item interaction data.

By achieving these objectives, we aim to advance the field of recommender systems by incorporating personality-based approaches, overcoming data collection challenges, and improving recommendation accuracy in diverse domains.

3. Applied theories and methods

In this section, we address the theories from psychological studies that describe how personality traits influence the choices of a person and how social connections, similarities and trust can be built upon the personality features of individuals. We also discuss the Big Five personality, which is one of the most studied personality models in the field of personality-based recommender systems. Lastly, we present the machine learning methods and models relevant to our proposed framework. Table 2 presents the notations used for the rest of the paper.

Table 1
Summary of literature review

Author	Year	Collection of personality scores	Cross domain framework	Similarity between users
Hu and Pu (2011)	2011	Survey	–	–
Khan et al. (2020)	2020	Social media posts	–	–
Aguiar et al. (2020)	2020	Reviews	–	–
Yakhchi et al. (2020)	2020	Social media posts	Common users	Nearest neighbours
Khodabandehlou et al. (2021)	2021	Questionnaire	–	EKNN algorithm
Wang et al. (2021)	2021	Reviews	Knowledge Transfer	–
Polignano et al. (2021)	2021	Social media posts	–	cosine similarity
Martijn et al. (2022)	2022	Crowd based study	–	–
Abolghasemi et al. (2022)	2022	Survey and questionnaire	–	Pairwise influence
Christodoulou et al. (2022)	2022	Reviews	–	–
Wang et al. (2022)	2022	Reviews	–	–
Soundariya et al. (2022)	2022	Reviews	Deep Neural network	–
Vatani et al. (2023)	2023	Reviews	–	Pearson's Correlation

Table 2
Summary of notations used in the model.

Notation	Interpretation
m	Number of items
n	Number of users
$R^{n \times m}$	User-item rating matrix
k	Latent space dimension
$U^{n \times k}$	User latent matrix
$I^{m \times k}$	Item latent matrix
$\mathcal{U}$	Set of users $\mathcal{U} = \{u_1, u_2, \dots, u_n\}$
$\mathcal{I}$	Set of items $\mathcal{I} = \{i_1, i_2, \dots, i_m\}$
N_u	Neighbourhood set of user u
$U^{I(N_u) \times k}$	Updated latent features matrix of user u along with its neighbours
$\mathbb{I}$	Indicator matrix
$\mathcal{N}$	Normal distribution
σ_U	Variance of user latent features
σ_I	Variance of item latent features
σ_R	Variance of rating matrix
σ_S	Variance of latent features of the neighbourhood
D^{source}	Source Domain
D^{target}	Target Domain
P_u	Personality features of user u
$SimPers_{u,v}$	Personality wise similarity score between user u and v
$SimRate_{u,v}$	Rating wise similarity score between user u and v
$S_{u,v}$	Overall similarity score between user u and v
α	Learning rate for gradient descent
β	Ratio of personality-wise similarity in overall similarity score
γ	Threshold constant for neighbourhood computation
λ_I	Regularization parameter for items
λ_S	Regularization parameter for neighbours
λ_U	Regularization parameter for users

3.1. Theories

Personality is the combination of individual characteristics that sets an individual's behavioural, emotional and cognitive aspects under different environments and contexts. Among multiple models and standards, computing personality like the HEXACO model (Ashton, Lee, & De Vries, 2014), and the Myers-Briggs Type Indicator model (largely known as MBTI model) (Boyle, 1995), the most widely recognized in studies is the Five-factor model (Goldberg, 1990), also known as the Big Five model. In the next section, we delineate the Big Five personality model and various approaches to obtaining the Big Five personality scores.

3.1.1. Personality factors and big five personalities

The Big Five personality model developed by Goldberg (1990) categorizes the personality profile into 5 following traits, namely Openness (O), Conscientiousness (C), Extraversion (E), Agreeableness (A), and Neuroticism (N), together referred to as OCEAN.

1. Openness: This characteristic describes one's imagination, creativity and insights. People with high openness have a broad range of interests, with great zeal and eagerness to learn and experience new things. Persons with low openness values are

seen to follow a monotonous approach to life and try to stay in their comfort zones.

2. Conscientiousness: The trait is responsible for showing thoughtfulness, control over impulses and decision-making capabilities. Individuals with high conscientiousness often show organized and structured behaviour, they plan early, make thoughtful decisions and analyse their behaviour and actions to observe how they affect others.
3. Extraversion: The trait defines how one person behaves in a company or gathering of many other people. Individuals with high extraversion show high emotions and interactions, while those with low extraversion prefer fewer social interactions and seldom express high emotions.
4. Agreeableness: This trait shows kindness, affection and trust. Any person with a high agreeableness score shows more kindness, empathy and affection than someone with a low agreeableness score.
5. Neuroticism: The characteristic describes a person's emotions and social behaviours. It demonstrates the physical and emotional response and impulses of a person. Anxiety and emotional balance are two major characteristics of neuroticism. Individuals with higher neuroticism traits experience anxiety, moodiness,

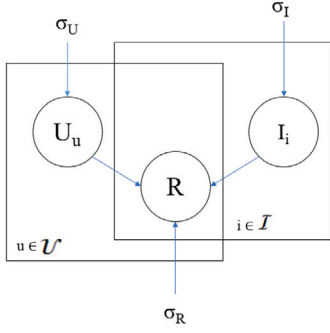


Fig. 1. Platonic representation of probabilistic matrix factorization.

and overthinking a lot. While people who have lower scores of neuroticism show a much more stable and resilient attitude.

From the literature on psychology and social sciences, it was proven that personality plays a big role in determining a person's interests. Kassarijan (1971) showed the presence of a correlation between the personality of users and their preferences; and how that can be used in analysing consumer behaviour. Finn (1997) showed how individuals with a higher degree of one personality trait show significant differences in a book or movie choice than individuals having a lower degree of the same trait. Subsequently, incorporating the personality aspects of the users into the recommendation model would help bring about better and more accurate personalized results. Personality-based attributes, along with rating information, could be effectively incorporated in computing the similarity of the users as well as the relevance of specific items for a user. For effective implementation of such factors, they could be amalgamated in a model-based approach as well. It is also seen that a user's interests can be analysed in a better way by considering other similar users impacting each other's recommendations. Works by Friedkin (1998) and Youyou et al. (2017) stated that users with similar personalities have similar preferences towards items. The similarity between users' rating behaviour or auxiliary attributes can help determine the interests of a user with more efficiency, which can be viewed as a social trust or friendship between the two users.

3.2. Methods

3.2.1. Probabilistic matrix factorization

A recommender system considers a rating matrix $R_{u,i}^{n \times m}$ containing the user-item interactions of n users from $\mathcal{U} : \{u_1, u_2, \dots, u_n\}$ and m items from $\mathcal{I} : \{i_1, i_2, \dots, i_m\}$. The rating $R_{u,i}^{n \times m}$ given by user u on item i is normally expressed in a range of 1 to 5. The matrix factorization technique factorizes the rating matrix into a user latent matrix $U^{n \times k}$ and item latent matrix $I^{m \times k}$, where k is the number of latent dimensions. Probabilistic matrix factorization is a widely used matrix factorization method, which assumes the latent features of users and items to be Gaussian distributed. Fig. 1 gives the platonic representation of probabilistic matrix factorization.

The rating matrix $R^{n \times m}$ having a mean μ_R and variance σ_R^2 can be represented as

$$p(R|U, I, \sigma_R^2) = \prod_{u=1}^n \prod_{i=1}^m [\mathcal{N}(R_{u,i} | g(U_u I_i^T), \sigma_R^2)]^{\mathbb{I}_{u,i}} \quad (1)$$

Here $\mathcal{N}(x|\mu, \sigma^2)$ is the normal distribution of the ratings and $\mathbb{I}_{u,i}$ is the indicator matrix, $\mathbb{I}_{u,i} = 1$ if $R_{u,i} \neq 0$, and 0 otherwise. The Normal priors for the user and item latent features are given as follows:

$$p(U|\sigma_U^2) = \prod_{u=1}^n \mathcal{N}(U_u | 0, \sigma_U^2 \mathbb{I}); \quad p(I|\sigma_I^2) = \prod_{i=1}^m \mathcal{N}(I_i | 0, \sigma_I^2 \mathbb{I}) \quad (2)$$

Using the Bayesian inference, the posterior probability of the latent features can be computed as Eq. (3).

$$p(U, I | R, \sigma_R^2, \sigma_U^2, \sigma_I^2) \propto p(R|U, I, \sigma_R^2) p(U|\sigma_U^2) p(I|\sigma_I^2) = \prod_{u=1}^n \prod_{i=1}^m [\mathcal{N}(R_{u,i} | U_u I_i^T, \sigma_R^2)]^{\mathbb{I}_{u,i}} \\ \times \prod_{u=1}^n \mathcal{N}(U_u | 0, \sigma_U^2 \mathbb{I}) \times \prod_{i=1}^m \mathcal{N}(I_i | 0, \sigma_I^2 \mathbb{I}) \quad (3)$$

Maximizing the log posterior over the user and item latent features and by the minimization of the following objective function (Eq. (4)), we obtain the sum of squared errors with the quadratic regularization terms.

$$L(R, U, I) = \frac{1}{2} \sum_{u=1}^n \sum_{i=1}^m \mathbb{I}_{u,i}^R (R_{u,i} - g(U_u I_i^T))^2 + \frac{\lambda_U}{2} \sum_{u=1}^n U_u^T U_u + \frac{\lambda_I}{2} \sum_{i=1}^m I_i^T I_i \quad (4)$$

3.2.2. Cross domain architecture

In a cross-domain recommendation system (Zang, Zhu, Liu, Zhang, & Yu, 2022), the objective is learning the ratings in one domain by exploiting information available in one or multiple domains by establishing a relationship between them. The domain in which the ratings are to be predicted is known as the target domain D^{target} , and all the other domains from which information is accessed for training the model are called the source domains D^{source} . For effective learning, the source and the target domain must be related. One trivial approach to create the relationship is with the help of common or overlapping users and items between the two domains (Fernández-Tobías et al., 2016); however, such mappings are very rare. Another approach is through the transfer of the rating pattern from the source domain to the target domain using matrix factorization (Ahmed, Saleem, Khalid, & Rashid, 2021; Zhang, Kong, & Zhang, 2023). This process is based on orthogonal non-negative matrix tri-factorization (NMTF), and is widely used in cross-domain recommenders to reduce data sparsity problems. In this approach, the source matrix is factorized into three parts: two latent matrices for users and items and another third matrix, which is constructed by co-clustering of the source domain rating matrix indicating the ratings provided by any user cluster on any item cluster. The third matrix is then used in learning the missing values in the target domain through optimization. Although this model addresses data sparsity issues, there exists a compatibility problem when applied across domains. Domain adaptation techniques (Saunders, 2022) can mitigate the compatibility issue over the source and target domains. In recent years, transfer learning method (Torrey & Shavlik, 2010) by domain adaptation (Wang et al., 2021; Wu, Xue, Hou & Liu, 2022; Yu et al., 2022) has gained popularity to learn parameters across domains, which does not require any common users.

Transfer Learning (Torrey & Shavlik, 2010) is defined as learning the missing values in one domain by using the information of another domain. Given one or more source domains D^{source} with corresponding feature space x^{source} and label space y^{source} , and one target domain D^{target} with corresponding feature space x^{target} and unknown label space y^{target} , transfer learning enables predicting the label space y^{target} from the feature space x^{target} represented by a learning task $T^{target} = \{y^{target} | f(x^{target})\}$ using the knowledge of source domain, i.e., mapping of y^{source} and x^{source} represented by the corresponding learning task, $T^{source} = \{y^{source} | f(x^{source})\}$; where f is the objective predictive function defined as $f : (x \rightarrow y)$, $D^{source} \neq D^{target}$ and $T^{source} \neq T^{target}$. It is mostly done by training the model with both target domain feature space and source domain feature space, then predicting the target domain's label space.

3.2.3. Mixing strategies

While designing the cross-domain algorithm using transfer learning, the goal is to train the model on multiple datasets simultaneously for effective learning. In this process, the model's parameters are trained with the source domain feature space. However, this may lead to over-training or under-training if there is no adequate relationship between the instances from the two domains, resulting in noise in learning

and transferring imperfect knowledge to the target domain, popularly known as Negative Transfer (Raffel et al., 2020). This can be mitigated by finding a correlation factor between the two domains. Instances from source and target domains having similarity can be merged and given as input to the model to train with related instances from both domains, thus reducing negative transfer and improving performance (Raffel et al., 2020). Finding similar instances between source and target domains and merging them can be done using multiple strategies, popularly known as mixing strategies. One approach is grouping all the instances of source and target domains together (Wang et al., 2021), known as equal mixing. However, this may fail to reduce negative transfer leading to quick overfitting. On the other hand, the instances can be merged proportionately based on the degree of correlation between the datasets (Wu et al., 2022), known as proportional mixing. This strategy often poses limitations in the case of small datasets when sampling in proportion to the dataset's size may result in the model ignoring most of the data leading to underfitting. A better approach of adapting the domains is by considering a parameter that determines the correlation between the instances from both domains and merging them based on the correlation factor (Ahmed et al., 2021; Wu, Xue, et al., 2022; Yu et al., 2022). The parameter can be configured to ensure that the model is trained on instances from the source dataset sufficiently and has enough correlation with target dataset instances.

3.2.4. Transfer learning using transformers

To create a personality-based cross-domain recommendation system using transfer learning of information, it is essential to create a prediction model to predict personality scores. Transformers in this regard, can serve as an effective tool to predict the scores. Transformers are attention-based encoder-decoder-type architecture. The encoder has encoding layers, each of which maps the input sequence into an abstract and continuous representation that holds the information on which portions from the input have relevance with each other and passes the output to the next encoder layer.

Inside the encoder, attention is calculated between every token and produces embeddings in the sequence that contain information about the token itself and other relevant tokens. Three vectors are created simultaneously for each attention unit, Query(Q), Key(K), and Value(V) and the transformer model learns three weight matrices called Attention Heads, i.e., Query weights W_Q , Key weights W_K , and Value weights W_V . For each token, the input word embedding is multiplied by the weight matrices followed by getting a dot product between the query matrix and key matrix of the two tokens normalized by the dimension of the key vectors d_k . A softmax layer is applied at last as shown in Eq. (5).

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (5)$$

Each layer of the transformer has multiple attention heads. With multiple attention heads, the model creates multiple representations of the tokens relevant to each other. The multi-headed attention output does computation in parallel, whose output is normalized and concatenated as shown in Eq. (6), followed by passing into a feedforward neural network. The output is again added to another neural network and is normalized.

$$\text{Multiheaded Attention}(Q, K, V) = \text{Concat}(\text{Attention}(QW_i^Q, KW_i^K, VW_i^V)W^O) \quad (6)$$

where the matrices QW_i^Q , KW_i^K , and VW_i^V are the projection matrices that belong to individual attention head i and W^O is the final projection matrix belonging to the whole multi-headed attention head.

The decoder contains decoding layers that take the output from the encoder and generate an output sequence using contextual information. Each decoder contains two multi-headed attention layers and a feedforward network. The input goes through the embedding layer and the first multi-headed attention layer to get the attention score. The

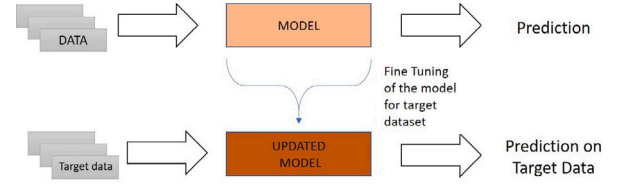


Fig. 2. Transfer Learning performed with transformers.

encoder's input is matched to the decoder's input by this process, which allows them to decide which input is relevant. The output is passed through a feedforward neural network for processing. To predict a numerical value for regression, a dense pooling layer and a dense linear layer are associated at the final stage of the architecture, and layers and parameters are updated as well so that they match the range of label space. Consequently, to perform transfer learning with the model, the parameters can be fine-tuned to learn on the target dataset as shown in Fig. 2.

4. Solution details

In this section, we discuss our proposed solution in two phases: first, we present a personality-based recommender model that utilizes the personality-wise similarity between users in a neighbourhood context, and then we delineate the cross-domain architecture that enables predicting personality scores of users in one domain by utilizing the review pattern and personality based information available in another domain.

4.1. Personality aware prediction model

4.1.1. Neighbourhood computation

We model our solution where we use the personality-wise similarity between the users to analyse their preferences owing to the theories suggesting that users having similar personality traits will possess similar preferences towards items. In our model, the personality features of the users are represented using $P\langle o, c, e, a, n \rangle$ where o, c, e, a and n refers to five traits of Big Five personality model (Goldberg, 1990), thereafter we use P instead of $P\langle o, c, e, a, n \rangle$ for ease of representation. Motivated by extant literature (Finn, 1997; Friedkin, 1998; Kassarian, 1971; Youyou et al., 2017), in the proposed model we exploited the influence of personality on user choices and the property that users with similar personality features will have similar item preferences. In this context, we envision that users' preferences are influenced by other users similar to the focal user. Thus, the features of the neighbours to the focal user should be incorporated along with the focal user's features. At first, we calculate the cosine similarity $\text{SimPers}_{u,v}$ between users u and v on personality trait scores to identify the neighbour users with similar personalities.

$$\text{SimPers}_{u,v} = \frac{P_u \cdot P_v}{|P_u| \cdot |P_v|} \quad (7)$$

where P_u and P_v denotes the personality feature vector of user u and v , respectively. We define a neighbourhood set N_u for each focal user u as follows:

$$N_u = \{v \in \mathcal{U} \mid \text{SimPers}_{u,v} \geq \gamma\} \quad \text{where } \gamma \in (0, 1) \quad (8)$$

To introduce the rating pattern similarity between the users, for all neighbours of u in N_u , we further calculate the cosine similarity between the users, $\text{SimRate}_{u,v}$, based on rating patterns as:

$$\text{SimRate}_{u,v} = \frac{R_u \cdot R_v}{|R_u| \cdot |R_v|} \quad \forall v \in N_u \quad (9)$$

The personality aspects and user rating patterns are further combined to calculate the overall similarity score between user u and v where $v \in N_u$. We combined personality-wise and rating pattern similarity to illustrate how personality factors would affect if assimilated with a collaborative filtering approach that takes only rating patterns into the picture. We introduce a parameter $\beta \in [0, 1]$ that decides the personality aspects' contribution to the overall similarity computation.

$$S_{u,v} = \beta \times \text{SimPers}_{u,v} + (1 - \beta) \times \text{SimRate}_{u,v} \quad (10)$$

Owing to the theory that users inheriting similar personality scores will possess similar item choices, the behaviour of user u will influence the neighbour users N_u . Hence, the feature vector of user u is updated using the latent features of the neighbours as

$$U' = \sum_{v \in N_u} S_{u,v} U_v \quad (11)$$

Here U' is the updated latent feature vector of user u with the latent feature vector of his neighbours, and $S_{u,v}$ is a value between $[0, 1]$ that represents the similarity between the two users. Herein, we posit that the behaviour of the users is directly affected by the neighbours, and therefore, we update the latent features with the similarity score $S_{u,v}$ and calibrate the effect of the neighbours. The higher the value of $S_{u,v}$, the higher the impact of the neighbour. The equation captures the latent feature vector of the users as the weighted average of the latent feature vector of its neighbours.

4.1.2. Personality enabled matrix factorization model

Our proposed model, **Personality Enabled Matrix Factorization**, PEMF assumes that similar users should have similar item preferences. So to produce an impact of similar users towards item prediction, we incorporate the neighbourhood users into the PMF model (Mnih & Salakhutdinov, 2007) by considering the neighbours' latent features and the user's latent features into the matrix factorization model.

$$p(U|S, \sigma_U^2, \sigma_S^2) = \prod_{u=1}^n \mathcal{N}(U_u|0, \sigma_U^2 \mathbb{I}) \times \prod_{u=1}^n \mathcal{N}(U_u | \sum_{v \in N_u} S_{u,v} U_v, \sigma_S^2 \mathbb{I}) \quad (12)$$

Eq. (12) is a normal distribution of the user's latent factors over the latent factors of its neighbours. Using Bayesian inference, the posterior probability of the user and item latent features is computed as follows:

$$\begin{aligned} p(U, I|R, S, \sigma_R^2, \sigma_S^2, \sigma_U^2, \sigma_I^2) &\propto p(R|U, I, \sigma_R^2) p(U|S, \sigma_U^2, \sigma_S^2) p(I|\sigma_I^2) \\ &= \prod_{u=1}^n \prod_{i=1}^m [\mathcal{N}(R_{u,i} | g(U_u I_i^T), \sigma_R^2)] \times \prod_{u=1}^n \mathcal{N}(U_u | \sum_{v \in N_u} S_{u,v} U_v, \sigma_S^2 \mathbb{I}) \\ &\quad \times \prod_{u=1}^n \mathcal{N}(U_u | 0, \sigma_U^2 \mathbb{I}) \times \prod_{i=1}^m \mathcal{N}(I_i | 0, \sigma_I^2 \mathbb{I}) \end{aligned} \quad (13)$$

The log of the posterior probability is calculated as follows:

$$\begin{aligned} \ln p(U, I|R, S, \sigma_R^2, \sigma_S^2, \sigma_U^2, \sigma_I^2) &= -\frac{1}{2\sigma_R^2} \sum_{u=1}^n \sum_{i=1}^m \mathbb{I}(R_{u,i} - g(U_u I_i^T))^2 - \frac{1}{2\sigma_U^2} \sum_{u=1}^n U_u^T U_u \\ &\quad - \frac{1}{2\sigma_S^2} \sum_{u=1}^n I_i^T I_i - \frac{1}{2\sigma_S^2} \sum_{u=1}^n ((U_u - \sum_{v \in N_u} S_{u,v} U_v)^T (U_u - \sum_{v \in N_u} S_{u,v} U_v)) \\ &\quad - \frac{1}{2} (\sum_{u=1}^n \sum_{i=1}^m \ln \sigma_R^2 - \frac{1}{2} ((n \times k) \ln \sigma_U^2 + (m \times k) \ln \sigma_I^2 + (n \times k) \ln \sigma_S^2) + c \end{aligned} \quad (14)$$

We maximize the log posterior over the user and item latent features, which is equivalent to the minimization of the following objective function:

$$L(R, S, U, I) = \frac{1}{2} \sum_{u=1}^n \sum_{i=1}^m \mathbb{I}(R_{u,i} - g(U_u I_i^T))^2 + \frac{\lambda_U}{2} \sum_{u=1}^n U_u^T U_u + \frac{\lambda_I}{2} \sum_{i=1}^m I_i^T I_i \quad (15)$$

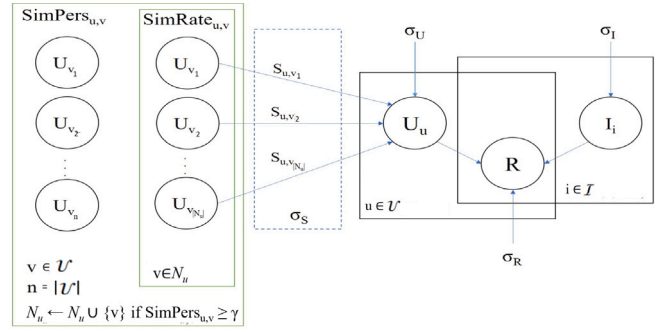


Fig. 3. Architecture of personality enabled matrix factorization model.

$$+ \frac{\lambda_S}{2} \sum_{u=1}^n ((U_u - \sum_{v \in N_u} S_{u,v} U_v)^T (U_u - \sum_{v \in N_u} S_{u,v} U_v))$$

Here, $\lambda_U = \frac{\sigma_U^2}{\sigma_R^2}$, $\lambda_I = \frac{\sigma_I^2}{\sigma_R^2}$, $\lambda_S = \frac{\sigma_S^2}{\sigma_R^2}$. The function $g(x)$ is modified sigmoid function used for bounding the range between $[1, 5]$ where

$$g(x) = \min_{u \in U', i \in I} (\{R_{u,i}\}) + \frac{\max_{u \in U', i \in I} (\{R_{u,i}\}) - \min_{u \in U', i \in I} (\{R_{u,i}\})}{1 + e^{-x}}$$

We can find the local minimum of the objective function by a gradient descent over U and I for all the users u and the items i as given in Eqs. (16) and (17).

$$\begin{aligned} \frac{\partial L}{\partial U_u} &= \sum_{i=1}^m \mathbb{I}_{u,i}^R I_i g'(U_u I_i^T) (g(U_u I_i^T) - R_{u,i}) + \lambda_U U_u \\ &\quad + \lambda_S (U_u - \sum_{v \in N_u} S_{u,v} U_v) - \lambda_S \sum_{(v|u \in N_v)} S_{v,u} (U_v - \sum_{w \in N_v} S_{v,w} U_w) \end{aligned} \quad (16)$$

$$\frac{\partial L}{\partial I_i} = \sum_{u=1}^n \mathbb{I}_{u,i}^R U_u g'(U_u I_i^T) (g(U_u I_i^T) - R_{u,i}) + \lambda_I I_i \quad (17)$$

The initial values U and I would be instances from a normal noise with zero mean. During learning, in every iteration, U and I would be updated based on the latent values from its previous iteration. Fig. 3 gives a plate notation of the model.

4.2. Cross domain framework

In the cross-domain framework, the objective is to learn the ratings in the target domain by exploiting information available in the source domain. In the proposed approach, **Personality Enabled Matrix Factorization for Cross Domain**, PEMF-CD, we require the user personality scores in the target domain. The personality scores of the target domain users are extracted through transfer learning from the source domain. To achieve this goal, user-generated reviews have been used from which personality can be predicted (Cachero et al., 2022; Christodoulou et al., 2022; Vatani et al., 2023). We concatenate the reviews from the source domain and the target domain by finding similar reviews. Merging all the reviews from both would induce unwanted noise in the corpus leading to negative transfer and ineffective learning as reviews in the source domain have varied correlations with reviews in the target domain. The algorithmic description of the prediction of personality scores from reviews across multiple domains is given in Algorithm 1.

4.2.1. Concatenating reviews from source and target domains

To avoid negative transfer, we have to adopt a mixing strategy of combining similar instances from the source and target domain. For this purpose, at first, we cluster the reviews from the source domain dataset into z clusters. This would ensure similar reviews come together as in step 3 of the algorithm. Next, we pick a review instance from the target

Algorithm 1 Prediction of personality from text using mixing strategy and joint embedding

```

1: Input:  $R^{source}, R^{target}, p^{source}$ 
2: Output:  $p^{target}$ 
3: Use K-Means to cluster  $R^{source}$  reviews into  $z$  clusters  $\{R_1^{source}, R_2^{source}, \dots, R_z^{source}\}$ 
4: for each  $r^{target} \in R^{target}$  do
5:   for each  $z$  do
6:      $cosineSim \leftarrow 0$ 
7:     for each  $r_z^{source} \in R_z^{source}$  do
8:        $cosineSim = cosineSim + cosineSimilarity(r^{target}, r_z^{source})$ 
9:      $L_z = cosineSim / |R_z^{source}|$ 
10:     $c = \arg\max(\{L_1, L_2, \dots, L_z\})$ 
11:     $R_c^{source} \leftarrow R_c^{source} \cup \{r^{target}\}$ 
12: //Cluster  $z$  contains reviews from both  $R^{source}, R^{target}$ 
13: for each  $z$  do
14:   Train Test Split( $z$ ) s.t. Train set  $\leftarrow \{R_z^{source}, p_z^{source}\}$ , Test set  $\leftarrow R_z^{target}$ 
15:   for each  $p_z^{source} \in p_z^{source} < o, c, e, a, n >$  do
16:     Initialise and Train  $BERT_{base}(R_z^{source}, p_z^{source})$ 
17:      $p_z^{target} \leftarrow \text{Test } BERT_{base}(R_z^{target})$ 
18:  $p^{target} \leftarrow \bigcup_z p_z^{target}$ 

```

domain and compute its cosine similarity with every instance within a source domain cluster (steps 4 to 8). We take the average of all the cosine similarities computed in L_z to represent the overall similarity of the cluster z with the target domain review in step 9. We repeat these steps for all the clusters of the source domain to identify the cluster with the maximum cosine similarity and attach the target domain review to that cluster R_c^{source} in steps 10 and 11. We follow this for all instances in the target domain and finally obtain multiple sets of reviews, each having instances from both the source and target domains. This process would prevent noise and negative transfer as similar instances would be trained together.

4.2.2. Joint embedding of source and target domain data and prediction

After the reviews from the source and target domains were merged, we performed a joint embedding of the texts, creating an embedding space representing characteristics from both domains. This way, a relationship between domains is established to enable transfer learning of personality scores. We have employed Bidirectional Encoder Representations from Transformers a.k.a BERT transformer model (Devlin, Chang, Lee, & Toutanova, 2018) to perform word embedding and predict personality scores.

BERT is a language model that has been widely used in the literature for downstream tasks in NLP (Choi, Kim, Joe, & Gwon, 2021; Koroteev, 2021). BERT is composed of transformer encoder layers. Using BERT, we learn the latent representations of text, which help to predict the Big Five personality scores. BERT applies bidirectional training of the transformer to language modelling. The text is put directly in sequence, looking at the text sequence from left to right and in opposite directions gives a deeper sense of language context and flow. While a transformer includes an encoder for taking the sequence in and a decoder for making predictions, BERT only uses the encoder part for embedding. After generating embeddings for the prediction task, a dense pooling layer is attached to the architecture of BERT. A dense linear layer with dropout would be the last final layer, such that BERT, which primarily involves classification, can predict numerical values, and the personality scores will be given as output.

Next, in steps 13 to 17, for each cluster z , we employ one BERT model that processes the text for joint word embedding and trains itself on the embedding space of reviews from the source dataset. The embedding is done jointly on the text from both domains, while the training of the model is done only on the source domain data. We split the reviews such that source domain reviews comprise the training set, and target domain reviews come under the test set. After the model is trained, we predict the personality scores of users belonging to the target domain by applying the model to the test set. We repeat this

process for all clusters and combine all the results to finally obtain the personality scores for all users in the target domain in step 18. Over the entire process, it is essential to maintain a validation set over the entire process to ensure that the models are not getting overfitted. The architecture of predicting the personality scores using mixing strategy and BERT is shown in Fig. 4.

Once we acquire the personality scores for the target dataset as well, they are used in the PEMF to predict items based on personality.

5. Dataset description

We conduct the experiments in two phases: firstly, we validate the proposed PEMF, and secondly, we integrate PEMF into a cross-domain framework, PEMF-CD. A publicly available dataset from TripAdvisor.com (Roshchina et al., 2015) has been used to experimentally validate PEMF, whereas, for PEMF-CD, four datasets from Amazon (Ni et al., 2019) have been used.

1. TripAdvisor: The TripAdvisor dataset (Roshchina et al., 2015) contains users' ratings, along with users' textual reviews. Alongside, it provides the demographics of the users such as location, age, and gender of the users. The dataset contains 32,580 reviews by 1098 users on 26,386 points of interest (POI), of which there are 12,736 hotels, 9683 restaurants, and 3967 places of attraction across 5927 cities worldwide. The user base comprises 438 males(39.8%), 373 females(33.9%), and the rest(26.3%) are unknown. Users belong to five age groups(18–24, 25–34, 35–49, 50–64, 65+). The dataset also reports users' Big Five personality scores, ranging between 0 and 1. The dataset was originally constructed by crawling through the contents of the TripAdvisor website, and the Big Five personality scores were assigned to the users using the LIWC tool (Mairesse et al., 2007) by analysing the linguistic features of the user reviews of the hotels and restaurants from the dataset. A higher value of a personality trait indicates the dominance of that personality trait of the individual.
2. Amazon: We have considered four datasets from Amazon (Ni et al., 2019) containing user reviews from four product categories, namely, Digital Music, Fashion, Magazine Subscriptions, and Video Games, each containing user and item IDs with user reviews and corresponding ratings on a 1–5 scale.

To evaluate the cross-domain architecture, the four datasets from Amazon have been taken as the target domains, whereas the TripAdvisor dataset is considered as the source domain. Table 3 presents the dataset statistics. It is worth noting that the Amazon dataset does not contain the personality scores, and the transfer learning method is used to predict the personality scores and integrate them into the model for predicting ratings.

6. Experimental results

In this section, we evaluate the performance of the proposed models PEMF and PEMF-CD against the state-of-the-art baseline algorithms with the dataset details and the experimental setup.

For the experiments, we have considered only the users that have rated at least five items and only those items that have been rated at least five times for the Digital Music and Video Games dataset. For Fashion and Magazine Subscriptions datasets, we take the users and items rated at least twice and once, respectively. However, we must ensure that the model gets trained perfectly for predictions. If the item has not been rated, we have marked it as 0. In the cross-domain recommendation algorithm, the first task is to predict the personality scores for the users in Amazon datasets using transfer learning. To avoid negative transfer during learning, we adopted a mixing strategy of selecting similar reviews from both domains. At first, we preprocessed the

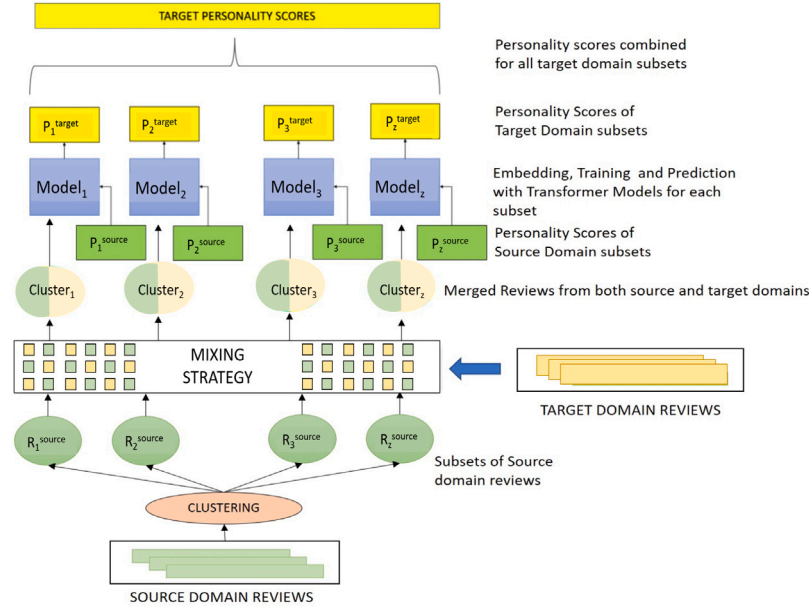


Fig. 4. Cross domain architecture.

Table 3
Description of the dataset.

Datasets	Number of users	Number of items	Number of reviews
TripAdvisor	1,098	26,386	32,580
Music	29,508	11,944	198,399
Fashion	19,366	4,915	48,912
Magazine	11,070	1,196	28,223
Video Games	44,187	14,863	426,791

reviews from all the datasets to remove the stop words, URLs, hashtags, etc, followed by tokenization and lemmatization. Word embeddings were generated on the text using Word2Vec. Next, we employed k-means clustering on the embedding space of source domain reviews and grouped the reviews into 6 clusters. The number of clusters was chosen by varying the cluster numbers and observing the best silhouette score. After we group the source domain reviews, we compute the cosine similarity scores of all the target domain reviews with every cluster of source reviews. The review from the target domain is then merged with that specific cluster that has the highest cosine similarity. In this way, we obtain sets of reviews from both source and target domains which are similar to each other such that negative transfer can be avoided.

After merging the reviews from source and target datasets, we use the BERT base model (Devlin et al., 2018) for the prediction of personality scores of target domain users. The BERT base model comprises 12 encoders with 12 bidirectional self-attention heads totalling 110 million parameters. Individual BERT models are created for every five personality traits and in every six clusters, making 30 BERT models to be trained for regression. Since BERT handles both embedding and prediction, hence we input the text directly into the model. The training is done on source domain reviews for the cluster, and personality scores are predicted for target domain reviews taken as test data. After obtaining the scores for all six clusters, they are merged to get the personality scores for all users in the target domain.

6.1. Experimental details

6.1.1. Experimental settings

All experiments were conducted on a Windows system with a 1.99 GHz Intel Core i3 processor and 16 GB RAM. The hyperparameters were tuned using a grid search algorithm and were set based on the best values obtained from pre-run experiments.

6.1.2. Evaluation metrics

To evaluate the efficacy of the proposed models, we have chosen two widely used accuracy measures as follows:

1. Root Mean Square Error(RMSE): RMSE is defined as the square root of the average residual errors between the actual values and predicted values from the model.

$$RMSE = \sqrt{\frac{\sum_{R_{u,i} \in \mathcal{T}} (R_{u,i} - \hat{R}_{u,i})^2}{|\mathcal{T}|}}$$

2. Mean Absolute Error(MAE): MAE is the mean of all the residual errors between the actual and the predicted values.

$$MAE = \frac{\sum_{R_{u,i} \in \mathcal{T}} |R_{u,i} - \hat{R}_{u,i}|}{|\mathcal{T}|}$$

Here, $\mathcal{T}$ denotes the set of ratings to be predicted, $\hat{R}_{u,i}$ and $R_{u,i}$ denote the predicted rating and actual rating for item i given by user u . The lower the values of MAE and RMSE, the better the performance of the model.

6.1.3. Baseline models

The twelve state-of-the-art baselines include three personality-based recommendation algorithms and nine designed for cross-domain recommendations out of which two are for cross-domain personality-based recommendation systems.

1. PMF (Mnih & Salakhutdinov, 2007): The PMF algorithm (Probabilistic Matrix Factorization) transforms a sparse rating matrix into two matrices that represent user and item latent feature matrices, which can be used to reconstruct the original rating matrix.
2. Personality-based CF (Hu & Pu, 2011): The authors incorporated personality aspects in the recommendation by computing similarity between two users based on personality and creating a

- linear equation and a cascading model to alleviate data sparsity issues.
3. LightFM (Kula, 2015): A hybrid recommendation system using a linear combination of user and item latent features and user and item metadata.
 4. LightFM-Pers: A modified version of LightFM has been implemented, including the user personality scores and other user metadata in the linear model of LightFM.
 5. APAR (Yakhchi et al., 2020): In this work, APAR (Analysis of Personality Aspect in Recommender system), the authors identified the personality trait scores of users using the LIWC tool from the reviews on Amazon. They proposed a cross-domain recommendation system integrating users' rating patterns, personality types, number of reviews, and ratings. In this work, they considered common users to map multiple domains; however, in practice, obtaining common users across domains is not always feasible.
 6. P²MF (Wang et al., 2021): A cross-domain personality-based recommendation system P²MF (Personality boosted Probabilistic Matrix Factorization), in which the Big Five personality scores for target domain users were predicted with five SVMs from user reviews and later used with PMF algorithm to generate recommendations. The authors in this work used the personality scores by feeding them into the PMF algorithm with a multiplicative factor that impacted the rating behaviour and, thus overall, the recommendations, which they referred to as Personality boosted PMF(P²MF).
 7. PTUPCDR (Zhu et al., 2022): A cross-domain recommender system PTUPCDR (Personalized Transfer of User Preferences for Cross-domain Recommendation) which creates a meta network of user embedding and employs a task-oriented optimization procedure with attention mechanism on overlapping and non-overlapping users to transform the embedding from source to target domain and utilizing the network to solve cold-start user problems in the target domain.
 8. CDRIB (Cao, Sheng, Cong, Liu, & Wang, 2022): This work CDRIB(Cross Domain Recommendation using Information Bottleneck) shares information on user-item interactions between the domains using IB regularizers. It captures the user-item interactions using a bipartite graph encoder and generates embedding for each domain. By analysing the cross-domain and intra-domain user-item interactions and user-user correlations across domains, the authors can generate the embedding for the users in each domain. In this case, the authors have considered both overlapping and non-overlapping users among the domains.
 9. UniCDR (Cao et al., 2023): The paper proposes a unified framework that can model different cross domain recommendation (CDR) scenarios by transferring the domain-shared domain specific information. UniCDR attempts to solve the data sparsity and cold start issues in CDR for overlapping users and items as well as intra and inter domain recommendations using contrastive loss.
 10. RealHNS (Ma et al., 2023): A novel model-agnostic framework is proposed that distinguishes the false from all the real hard negative samples (HNS) i.e. when any user-item interaction is not considered even if the item is observed by the user, by leveraging negative transfer in cross domain recommendations. A dynamic hard negative sampling filter is used for making the classification.
 11. NMCDR (Xu et al., 2023): The authors in this work (Node Matching Cross Domain Recommender) have designed the framework for partially overlapped users. It takes intra-domain knowledge fused with inter-domain knowledge with fully connected user-user graph and learns from the user-item matching information to predict all missing interactions.
 12. MAN (Lin et al., 2024): This work (Mixed Attention Network) uses attention mechanism to extract information from the same domain and cross-domain systems. An encoding layer captures the sequential patterns followed by attention layers of three kinds to group the users across different domains and predict the domain-specific and cross-domain interactions.

6.1.4. Evaluation of performance

Experiments have been conducted on datasets from TripAdvisor and Amazon to validate the efficacy of the two models PEMF and PEMF-CD, respectively. Both algorithms have been trained with two sets of training data, 80% and 90% of the datasets, keeping the rest of the data for testing. Each experiment is done 5 times and the average values are reported. For the proposed algorithm, we have taken the learning rate as α , regularization terms of user latent features (λ_U), item latent features (λ_I), neighbourhood latent features (λ_S), the number of latent features (k), similarity coefficient (β), and neighbourhood threshold parameter (γ). All the hyperparameters are tuned with a grid search algorithm to get the optimal values as reported in Table 4.

We evaluated our proposed approach PEMF and PEMF-CD and compared it to thirteen baselines, nine of which were specifically designed for cross-domain recommendations. Consequently, PEMF was compared to only four baselines. The results shown in Table 4 demonstrate that PEMF outperforms the four baseline algorithms, achieving improvements of up to 73.2% and 75.2% in RMSE and MAE, respectively, for a 90:10 train-test split. Similar improvements are observed for the 80:20 split, with enhancements of up to 72.6% and 74.7% in RMSE and MAE, respectively. $\beta \geq 0.5$ in PEMF leads to lower RMSE and MAE, indicating that personality features contribute more significantly than rating patterns to recommendations and play a crucial role in improving accuracy. For different train-test splits, the optimal γ values range from 0.6 to 0.8 when constructing neighbourhoods. This suggests that instead of considering all users, only those with high personality-wise similarity should be regarded as neighbours, thus reducing noise in the model.

From Tables 5, 6, 7 and 8 it is evident that PEMF-CD surpasses the baseline algorithms in cross-domain platforms with improvements of up to 24.72%, 64.28%, 48.79%, and 61% in RMSE, and 55.9%, 76.7%, 67.6%, and 71.5% in MAE for a 90:10 train-test split and 28.5%, 64.1%, 48.5% and 60.2% in RMSE and 57.7%, 76.3%, 64.7% and 63.3% in MAE for 80:20 train-test split with Digital Music, Fashion, Magazine Subscriptions and Video Games datasets from Amazon, respectively. These improvements can be attributed to two main features of our method: the careful incorporation of personality scores into the PMF technique and the strategic compilation of user reviews from both domains to learn the linguistic patterns of users with similar personality traits.

6.2. Comparison against baselines

Comparing the performance to other baselines, PMF and LightFM exhibit higher RMSE and MAE than PEMF. LightFM-Pers, a modified version of LightFM with personality scores, shows improved accuracy compared to LightFM, emphasizing the importance of personality scores in the model. In PEMF, instead of using the personality scores of a user directly in the model, it has been utilized to identify similar users by updating the latent features, resulting in superior performance. Even Personality-based CF and LightFM-Pers, which employ personality scores, do not update the user's latent features in the linear model, leading to inferior performance compared to PEMF. In the cross-domain recommender system P²MF, the personality scores for target domain users are treated as binary values, collected through classification using five independent SVMs for five personality traits. However, this approach fails to capture the true intensity of personality in the model. In

Table 4
Performance of algorithm on Tripadvisor dataset.

Test data		PMF	PCF	LightFM	LightFM-Pers	PEMF
10%	RMSE	16.187	24.912	16.173	13.719	6.665
	Improvement%	58.83%	73.25%	58.79%	51.42%	
	MAE	4.012	4.985	3.875	3.505	1.233
	Improvement%	69.27%	75.27%	68.18%	64.82%	
20%	RMSE	16.249	23.905	16.251	12.079	6.540
	Improvement%	59.75%	72.64%	59.76%	45.86%	
	MAE	4.025	4.791	3.888	3.279	1.208
	Improvement%	69.99%	74.79%	68.93%	63.16%	

Table 5
Performance of PEMF-CD on Amazon Music dataset.

Test data	10%				20%			
	RMSE	Improve%	MAE	Improve%	RMSE	Improve%	MAE	Improve%
P ² MF	11.683	−35.92%	2.526	−25.69%	11.348	−32.88%	2.464	−22.36%
APAR	21.093	24.72%	4.123	22.99%	21.093	28.51%	4.123	26.87%
PTUPCDR	10.051	−57.97%	3.214	1.23%	10.862	−38.81%	3.513	14.18%
CDRIB	10.133	−57.70%	3.282	3.28%	10.223	−47.49%	3.04	0.82%
UniCDR	12.875	18.90%	4.108	29.40%	18.045	19.70%	3.637	20.60%
RealHNS	11.082	−70.36%	3.200	0.78%	11.085	−36.03%	2.830	−6.54%
NMCDR	9.572	−97.23%	3.050	−4.10%	16.760	10.03%	3.178	5.13%
MAN	16.892	−11.76%	7.210	55.96%	16.932	10.94%	7.123	57.67%
PEMF-CD	18.879		3.175		15.079		3.015	

Table 6
Performance of PEMF-CD on Amazon Fashion dataset.

Test data	10%				20%			
	RMSE	Improve%	MAE	Improve%	RMSE	Improve%	MAE	Improve%
P ² MF	11.091	37.25%	2.538	45.16%	11.211	37.74%	2.542	45.08%
APAR	19.483	64.28%	3.712	62.50%	19.479	64.16%	3.712	62.39%
PTUPCDR	7.927	12.20%	1.411	1.36%	7.451	6.32%	1.703	18.06%
CDRIB	7.063	1.45%	1.437	3.19%	7.334	4.83%	1.451	3.79%
UniCDR	7.173	3.10%	1.45	4.20%	7.19	3.01%	1.982	42.00%
RealHNS	8.64	19.41%	1.520	8.42%	7.37	5.23%	1.435	2.72%
NMCDR	8.98	22.49%	3.215	56.70%	8.52	18.10%	2.051	31.94%
MAN	10.43	33.27%	5.984	76.74%	11.02	36.63%	5.885	76.28%
PEMF-CD	6.96		1.392		6.98		1.396	

Table 7
Performance of PEMF-CD on Amazon Magazine Subscriptions dataset.

Test data	10%				20%			
	RMSE	Improve%	MAE	Improve%	RMSE	Improve%	MAE	Improve%
P ² MF	10.547	9.29%	2.515	23.92%	10.59	9.13%	2.488	22.65%
APAR	18.683	48.79%	3.444	44.44%	18.687	48.51%	3.446	44.15%
PTUPCDR	5.112	−87.11%	3.45	44.56%	5.101	−88.63%	4.406	56.31%
CDRIB	11.297	15.32%	3.306	42.13%	10.135	5.05%	3.289	41.48%
UniCDR	10.453	9.30%	1.985	3.80%	10.778	12.01%	2.204	14.50%
RealHNS	11.050	13.42%	5.120	62.64%	10.352	7.04%	4.042	52.38%
NMCDR	10.321	7.31%	2.032	5.86%	9.952	3.31%	2.165	11.09%
MAN	12.003	20.29%	5.910	67.63%	11.430	15.81%	5.456	64.72%
PEMF-CD	9.567		1.913		9.623		1.925	

Table 8
Performance of PEMF-CD on Amazon Video Games dataset.

Test data	10%				20%			
	RMSE	Improve%	MAE	Improve%	RMSE	Improve%	MAE	Improve%
P ² MF	11.159	30.64%	2.414	35.87%	11.093	28.81%	2.443	35.35%
APAR	19.845	61.00%	3.753	58.75%	19.879	60.27%	3.764	58.04%
PTUPCDR	11.450	32.40%	1.613	4.05%	11.436	30.94%	1.104	−42.91%
CDRIB	9.145	15.37%	1.872	17.33%	9.35	15.54%	1.436	−9.93%
UniCDR	9.776	26.30%	1.73	11.80%	10.374	35.90%	1.87	18.40%
RealHNS	8.82	12.24%	1.891	18.14%	8.240	4.16%	1.624	2.77%
NMCDR	9.51	18.61%	2.120	26.98%	9.036	12.61%	2.521	37.37%
MAN	10.66	27.37%	5.432	71.50%	10.326	23.52%	4.304	63.31%
PEMF-CD	7.74		1.548		7.897		1.579	

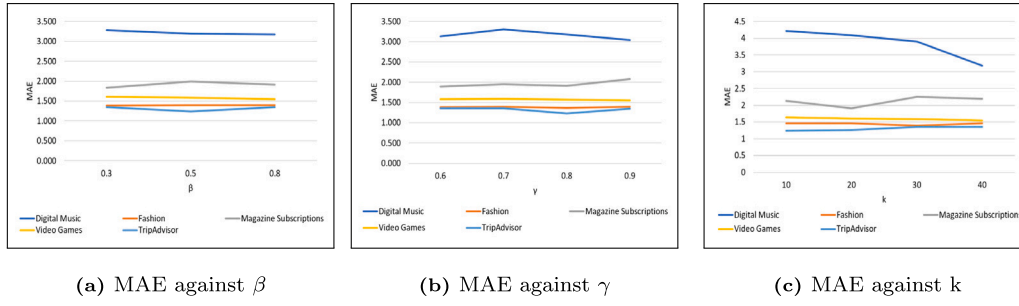


Fig. 5. Performance of PEMF-CD against different hyperparameters.

contrast, PEMF-CD employs real-valued personality scores and considers a neighbourhood of users with similar personalities who impact the recommendations of the focal user, resulting in a better performance compared to P²MF. Although APAR integrates the influence of similar users, it updates the rating pattern of users rather than the user's latent features, unlike PEMF-CD. PTUPCDR, CDRIB, UniCDR, RealHNS, NMCDR and MAN utilize the set of overlapping users to bridge between the source and target domain. In our experiments, to compare with the above-mentioned baselines, we have taken one domain as the target domain and the rest of the domains as source domains. In RealHNS, the optimization is achieved using transferring the information across domains like APAR. However, although it captures the user-item interaction information effectively, fails to capture the behavioural aspects of the users that can be obtained using inherent attributes of the users like personality. PTUPCDR and MAN employ a neural network method with attention. In PEMF-CD, while attention is already incorporated in the embedding and training model through using BERT, analysing text-based reviews to get personality-wise similar users have given more cross-domain information. CDRIB and NMCDR uses graph based technology to capture to transform the user-item interactions between the domains while UniCDR uses user's information from one domain combined with information transferred from other domains where the user has interacted. However, mutual information across the domains on user-item interactions proves to be very limited which might lead to inaccurate embedding impacting the model negatively. Unlike this, PEMF-CD exploits the linguistic cues of the reviews from the users of both domains to build the cross-domain model and ensure effective learning of the information.

The hyperparameters were tuned using a grid search algorithm and were set based on the best values obtained from pre-run experiments. The optimal values of the hyperparameters are given in Table 9.

7. Discussion and implications

The improvement in performance in PEMF and PEMF-CD can be attributed to using personality features in recommendations and creating a neighbourhood of users based on their similarity in personality to design a collaborative framework. The personality of the users, which shows a significant impact on their preferences, can henceforth be used in making improved recommendations than prior works. Also, the neighbourhood of the users based on personality-wise similarity plays an important role by influencing the latent features of the user towards getting recommendations. Unlike other works, which used the personality of the users directly in matrix factorization or updated the rating patterns of the users, our approach influences the latent features, which are responsible for generating a better performance. Further, to tackle the sparsity issue, the cross-domain framework PEMF-CD is proposed, where the personality scores are predicted by leveraging linguistic cues in user reviews using transfer learning through a transformer architecture. This as well shows an effective approach to collecting personality scores which overcomes the challenges of traditional survey-based methods. The experiments have been conducted for

both 80-20 and 90-10 split of train and test data; however, considering the space constraint, we have reported only the 90-10 split.

7.1. Impact of personality features

Prior literature (Finn, 1997; Kassarian, 1971) on psychology showed that personality plays a major role in determining a person's preferences. Consequently, incorporating personality features into a recommendation model shows a performance improvement. In this work, we have employed the computation of user similarity based on the Big Five personality traits. Additionally, we have included the similarity between users' rating patterns to establish an overall similarity score. The parameter β determines the effect of personality-wise similarity on the overall similarity score. Higher values of β indicate a higher contribution of personality-wise similarity compared to rating pattern-wise similarity. For analysing the impact of β on the performance of the model, β is varied with three values 0.3, 0.5, and 0.8, and the trend is shown in Fig. 5(a). It is seen that the algorithm performs optimally for a higher value of β . The findings are further supported by results as given in Table 4 and Tables 5, 6, 7 and 8; it is seen that prior works that did not use the personality features performed poorly in terms of accuracy. Hence, it can be inferred that personality, which plays a critical role in deciding preferences, can be used in recommender systems to improve their effectiveness.

7.2. Impact of neighbourhood

We create connections between a pair of users based on their personality similarities. Two users are considered neighbours if the similarity score exceeds a predefined threshold value γ . In the proposed approaches PEMF and PEMF-CD, along with the user's latent feature, the latent features of the neighbour users are also integrated. A higher value of γ considers only those users who have a significant amount of similarity in personality. Conversely, lower values of γ accumulate more users, adding insignificant influence and noise into the model, which can lead to underfitting. Our experiments suggest that both the models perform best when the value of γ is very high, thus validating the influence of selected neighbour users in the recommendations. In 5(b), we show the performance of PEMF-CD for different values of γ for a 90-10 train-test split.

7.3. Impact of latent factors

We performed our experiments on PEMF on TripAdvisor dataset and PEMF-CD on four Amazon datasets by varying the latent vector size k from 10 to 40. We observe that PEMF and PEMF-CD give the best results at different k values, which is shown in Fig. 5(c).

7.4. Ablation study

We have conducted an ablation study to understand the importance of various components of the proposed model. In PEMF-CD, the

Table 9
Hyperparameter distribution.

TripAdvisor	10%						20%					
	α	k	β	λ_U, λ_I	λ_S	γ	α	k	β	λ_U, λ_I	λ_S	γ
PMF	0.01	20	-	0.01	-	-	0.01	10	-	0.02	-	-
PCF	-	-	0.5	-	-	-	-	-	0.5	-	-	-
LightFM	-	40	-	-	-	-	-	40	-	-	-	-
PEMF	0.0001	10	0.5	0.04	0.8	0.8	0.0001	10	0.5	0.04	0.8	0.6
Music	10%						20%					
	α	k	β	λ_U, λ_I	λ_S	γ	α	k	β	λ_U, λ_I	λ_S	γ
PMF	0.01	10	-	0.01	-	-	0.01	10	-	0.01	-	-
PCF	-	-	0.5	-	-	-	-	-	0.6	-	-	-
LightFM	-	40	-	-	-	-	-	40	-	-	-	-
P ² MF	0.001	10	-	0.01	0.6	-	0.001	20	-	0.03	0.5	-
APAR	0.01	20	-	0.002	0.2	0.5	0.01	10	-	0.002	0.2	0.5
PTUPCDR	0.01	-	-	-	-	-	0.01	-	-	-	-	-
CDRIB	0.001	-	-	-	-	-	0.01	-	-	-	-	-
UniCDR	0.01	-	-	-	-	0.6	0.01	-	-	-	-	0.6
RealHNS	0.001	20	-	0.5	-	-	0.001	50	-	0.5	-	-
NMCDR	0.0001	50	0.5	1	1	-	0.0001	30	0.5	1	1	-
MAN	0.001	20	-	0.1	0.1	-	0.0001	20	-	0.3	0.3	-
PEMF-CD	0.001	40	0.8	0.03	0.8	0.8	0.001	20	0.5	0.03	0.1	0.6
Fashion	10%						20%					
	α	k	β	λ_U, λ_I	λ_S	γ	α	k	β	λ_U, λ_I	λ_S	γ
PMF	0.01	10	-	0.01	-	-	0.01	10	-	0.02	-	-
PCF	-	-	0.6	-	-	-	-	-	0.6	-	-	-
LightFM	-	40	-	-	-	-	-	40	-	-	-	-
P ² MF	0.001	30	-	0.03	0.3	-	0.001	30	-	0.01	0.3	-
APAR	0.01	10	-	0.001	0.1	0.5	0.01	10	-	0.002	0.1	0.5
PTUPCDR	0.001	-	-	-	-	-	0.001	-	-	-	-	-
CDRIB	0.01	-	-	-	-	-	0.01	-	-	-	-	-
UniCDR	0.001	-	-	-	-	0.7	0.001	-	-	-	-	0.7
RealHNS	0.001	20	-	0.5	-	-	0.001	50	-	0.2	-	-
NMCDR	0.0001	50	0.5	1	1	-	0.0001	50	0.5	1	1	-
MAN	0.01	20	-	0.3	0.3	-	0.0001	20	-	0.3	0.3	-
PEMF-CD	0.0001	30	0.5	0.04	0.1	0.9	0.01	30	0.5	0.04	0.1	0.9
Magazines	10%						20%					
	α	k	β	λ_U, λ_I	λ_S	γ	α	k	β	λ_U, λ_I	λ_S	γ
PMF	0.01	10	-	0.01	-	-	0.01	20	-	0.02	-	-
PCF	-	-	0.5	-	-	-	-	-	0.5	-	-	-
LightFM	-	40	-	-	-	-	-	40	-	-	-	-
P ² MF	0.001	20	-	0.02	0.7	-	0.001	10	-	0.02	0.8	-
APAR	0.01	20	-	0.002	0.3	0.5	0.01	20	-	0.003	0.5	0.5
PTUPCDR	0.01	-	-	-	-	-	0.01	-	-	-	-	-
CDRIB	0.1	-	-	-	-	-	0.001	-	-	-	-	-
UniCDR	0.001	-	-	-	-	0.7	0.001	-	-	-	-	0.7
RealHNS	0.001	50	-	0.5	-	-	0.001	100	-	0.5	-	-
NMCDR	0.0001	50	0.8	1	1	-	0.0001	50	0.5	1	1	-
MAN	0.001	40	-	0.5	0.5	-	0.001	20	-	0.1	0.1	-
PEMF-CD	0.0001	20	0.8	0.02	0.8	0.8	0.0001	30	0.8	0.02	0.3	0.8
Video Games	10%						20%					
	α	k	β	λ_U, λ_I	λ_S	γ	α	k	β	λ_U, λ_I	λ_S	γ
PMF	0.0001	40	-	0.01	-	-	0.0001	40	-	0.01	-	-
PCF	-	-	0.7	-	-	-	-	-	0.60	-	-	-
LightFM	-	40	-	-	-	-	-	40	-	-	-	-
P ² MF	0.001	30	-	0.01	0.7	-	0.01	30	-	0.02	0.6	-
APAR	0.01	30	-	0.003	0.3	0.6	0.01	20	-	0.001	0.5	0.6
PTUPCDR	0.01	-	-	-	-	-	0.01	-	-	-	-	-
CDRIB	0.001	-	-	-	-	-	0.01	-	-	-	-	-
UniCDR	0.001	-	-	-	-	0.6	0.001	-	-	-	-	0.6
RealHNS	0.001	20	-	0.2	-	-	0.001	50	-	0.2	-	-
NMCDR	0.0001	50	0.8	1	1	-	0.0001	50	0.8	1	1	-
MAN	0.0001	30	-	0.1	0.1	-	0.0001	50	-	0.1	0.1	-
PEMF-CD	0.0001	40	0.8	0.02	0.1	0.9	0.1	40	0.50	0.02	0.5	0.7

cross-domain framework is set up by connecting the domains through embedding of the text-based user reviews. The source domain reviews are clustered into subsets and each target domain review is mapped to one of the source domain subsets based on similarity criteria. Each of the subsets is then embedded together to learn and predict the personality scores. The prediction of personality scores is thus dependent upon how the reviews are mapped into subsets. Without this process, all the reviews could have been embedded and learned together with a model

to predict the personality scores. This strategy is called equal mixing. To show the efficacy of the mixing strategy we perform experiments with the equal mixing strategy and parameter scaled mixing applied in PEMF-CD. From the experiments, as shown in Fig. 6, it can be observed that the addition of the target domain reviews into subsets based on similarity criteria improves the final results of the recommendation up to 33.6% improvement in MAE. This can be attributed to the fact that training all the reviews together can lead to negative transfer

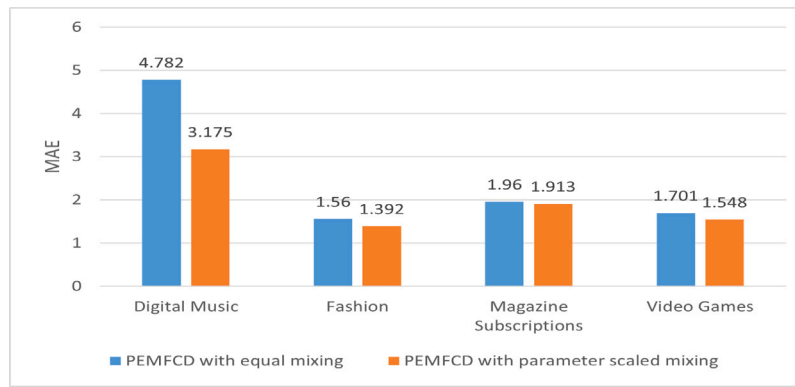


Fig. 6. Variation of MAE with different mixing strategies for 90:10 train test split.

as there is no adequate correlation between the instances from both domains. This negative transfer will induce unwanted noise in the model making inaccurate predictions of personality scores. On the other hand, the similarity factor between the source and target domain reviews ensures instances that are correlated get merged and embedded together reducing the negative transfer effect.

7.5. Implication of research

In this study, we introduced the concept of user neighbourhood based on personality and how the similarity between users can be leveraged to affect the recommendations. Also, we demonstrated how data sparsity issues would be handled using transfer learning in a cross-domain framework.

7.5.1. Using personality features for recommendations

Using personality features has been shown to improve the accuracy of recommendations already in prior works. In this work, combining personality-wise similarity and rating pattern similarity to affect the latent features of the user has been the key factor in improving the accuracy. Also, establishing the neighbourhood concept and considering other users with similar personalities who can influence each other's preferences is another key factor that improves the performance. The latent features of the neighbours have been used to modify the latent features of the user, thus putting more effect towards accurate and more personalized recommendations, which was not present in any of the earlier works. This approach shows how the personality of the users can be utilized to improve the performance of the recommendations and opens up new directions in the research of personality-based recommenders.

7.5.2. Handling data sparsity with transfer learning

This study also shows how personality features could be applied in cross-domain recommendations and tackle data sparsity issues. Since acquiring personality scores is difficult, our approach to predicting user personality scores from the text content of user-generated reviews using transfer learning is convenient and advantageous in several aspects. This also opens up the path for acquiring personality scores from multiple text-based sources by avoiding the challenges of survey-based methods. Also, it gives new research directions in developing cross-domain recommendations across related domains.

7.6. Implication of practice

One of the major limitations that any recommender system faces is data sparsity issues which leads to inferior quality or reduced accuracy. The transfer learning method of acquiring personality scores has shown how data sparsity issues can be reduced and effective recommendations can be predicted using personality features. These

findings have broader implications for natural language processing tasks beyond recommender systems. This way, it would help existing applications in generating recommendations for cold start users. Furthermore, since the personality features of a user act independently of the domain, they can be used in building cross-domain networks and can be learned simultaneously from other domains by transferring information. This would be very useful in analysing user profiles and user preferences across multiple domains. The use of a transformer model allows the algorithm to effectively capture underlying semantics and latent representations within textual data. This implies that the algorithm can leverage the nuanced meanings, writing styles, and context of user reviews, enabling more accurate predictions of personality traits. Additionally, the performance of the proposed algorithm, tested on several real-world datasets from different domains, and the significant improvement in recommendation accuracy, suggest that the algorithm is generalizable, making it applicable to diverse domains and datasets.

8. Conclusion

This study investigates the integration of user personality features into recommender systems and presents a novel approach for predicting user personality scores from reviews in the absence of explicit personality information. By leveraging transfer learning methods to establish cross-domain connections and mitigate data sparsity issues, this study explores the potential of incorporating user personality traits to enhance the accuracy and personalization of recommendations. Furthermore, the concept of the neighbourhood is introduced, wherein users with similar personalities influence each other's preferences, thereby refining the recommendation process. Through the modification of latent features based on the latent features of neighbours, more accurate and personalized recommendations are achieved. PEMF has been evaluated using a public dataset from TripAdvisor, which consists of users' ratings with their Big Five personality traits, showing a significant improvement over the baseline algorithms. We reconfigured our model and proposed PEMF-CD to apply in a cross-domain framework. Through transfer learning, we predicted the personality scores of Amazon users from their reviews using transfer learning by jointly embedding them with reviews from the TripAdvisor dataset. PEMF-CD also outperforms the baseline algorithms in RMSE and MAE. The experimental findings from both the algorithms, i.e., PEMF and PEMF-CD reveal that our proposed approaches have superior performance over the five state-of-the-art baseline cross-domain algorithms with improvements of up to 24.72%, 64.28%, 48.79%, and 61% in RMSE, and 55.9%, 76.7%, 67.6%, and 71.5% in MAE for a 90:10 train-test split and 28.5%, 64.1%, 48.5% and 60.2% in RMSE and 57.7%, 76.3%, 64.7% and 63.3% in MAE for 80:20 train-test split with Digital Music, Fashion, Magazine Subscriptions and Video Games datasets from Amazon, respectively.

8.1. Limitation and future work

In this study, we have incorporated Big Five personality traits extracted from text reviews along with the rating preferences, and the influence of other metadata about users or items has not been considered. The study does not address the full spectrum of user personality dimensions but rather focuses on five specific personality traits. However, personality being a complex and multidimensional construct, consideration of additional personality dimensions could potentially enhance the accuracy and personalization of recommendations. Furthermore, the study relies on the accuracy metrics and does not conduct any user study to explicitly explore user acceptance and adoption of personality-based recommendations. Users might have varying levels of trust in systems that employ personality information and thus, understanding user perceptions, concerns, and preferences regarding personalized recommendations based on personality traits is crucial for positive user engagement.

CRedit authorship contribution statement

Somdeep Acharyya: Conceptualization, Methodology, Data curation, Writing – original draft. **Nargis Pervin:** Supervision, Conceptualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The github repo is mentioned in the manuscript.

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